Tkinter

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1. What is Tkinter?

Tkinter is a built-in Python library that allows us to create **Graphical User Interfaces (GUIs)** such as windows, buttons, text boxes, and more.

You can make apps like calculators, forms, and notepads using Tkinter.

To use Tkinter, you simply import it:

import tkinter as tk

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2. Creating Your First Window

Every Tkinter program has three main steps:

1. Create a window
2. Set its title and size
3. Keep it running using mainloop()

```
import tkinter as tk


root = tk.Tk()              # Step 1: Create main window
root.title("My First App")  # Step 2: Set title
root.geometry("300x200")    # Optional: Set size (width x height)


root.mainloop()             # Step 3: Display window until closed
```

Explanation:

- Tk() creates the main application window.
- title() sets the window name.
- geometry() changes window size.
- mainloop() keeps the window active until the user closes it.

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3. Adding a Label (Text on Screen)

Labels display text or images inside the window.

```
import tkinter as tk
```

```
root = tk.Tk()
root.title("Label Example")

label = tk.Label(root, text="Welcome to Tkinter!")
label.pack()  # Adds label to the window

root.mainloop()
```

Explanation:

- Label() creates a label widget.
- pack() adds it to the window (places it automatically).

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4. Adding a Button

Buttons perform actions when clicked.

```
import tkinter as tk

def say_hello():
    print("Hello Students!")

root = tk.Tk()
root.title("Button Example")

button = tk.Button(root, text="Click Me", command=say_hello)
button.pack(pady=10)

root.mainloop()
```

Explanation:

- Button() creates a button.
- command=say_hello links the button to a function that runs when clicked.

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5. Taking Input from the User

Use an **Entry box** to take text input.

```
import tkinter as tk

def show_input():
    name = entry.get()  # Get text from entry box
    print("Hello", name)

root = tk.Tk()
root.title("Entry Example")

entry = tk.Entry(root)
entry.pack(pady=5)

button = tk.Button(root, text="Submit", command=show_input)
button.pack(pady=5)

root.mainloop()
```

Explanation:

- Entry() creates a single-line text box.
- .get() retrieves what the user typed.

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6. Layout Managers (Arranging Widgets)

Tkinter offers three ways to arrange widgets:

Method	**Description**	**Use**
pack()	Stacks widgets vertically or horizontally	Simple layouts
grid()	Arranges widgets in rows and columns	Forms and tables
place()	Positions widgets at exact x, y coordinates	Precise control

Example using **grid**:

```
import tkinter as tk

root = tk.Tk()
root.title("Grid Example")
```

```
tk.Label(root, text="Name:").grid(row=0, column=0)
tk.Entry(root).grid(row=0, column=1)

tk.Label(root, text="Age:").grid(row=1, column=0)
tk.Entry(root).grid(row=1, column=1)

root.mainloop()
```

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7. Message Boxes (Pop-ups)

Message boxes are small pop-up windows that display information or warnings.

```
import tkinter as tk
from tkinter import messagebox

def greet():
    messagebox.showinfo("Greeting", "Hello Everyone!")

root = tk.Tk()
root.title("Messagebox Example")

button = tk.Button(root, text="Click for Greeting", command=greet)
button.pack(pady=10)

root.mainloop()
```

Explanation:

- messagebox.showinfo() displays a pop-up with a title and message.

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8. Simple Calculator Example

This mini-project performs addition of two numbers.

```
import tkinter as tk
```

```
def add_numbers():
    n1 = int(entry1.get())
    n2 = int(entry2.get())
    result = n1 + n2
    result_label.config(text="Result: " + str(result))

root = tk.Tk()
root.title("Mini Calculator")

tk.Label(root, text="Enter First Number:").pack()
entry1 = tk.Entry(root)
entry1.pack()

tk.Label(root, text="Enter Second Number:").pack()
entry2 = tk.Entry(root)
entry2.pack()

tk.Button(root, text="Add", command=add_numbers).pack(pady=5)
result_label = tk.Label(root, text="Result: ")
result_label.pack()

root.mainloop()
```

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9. Changing Colors and Font

You can customize colors and fonts for widgets.

```
import tkinter as tk

root = tk.Tk()
root.title("Color Example")
root.configure(bg="lightblue")
```

```
label = tk.Label(root, text="Colorful Label", bg="yellow", fg="red", font=("Arial", 16))
label.pack(pady=10)

root.mainloop()
```

Explanation:

- bg → background color
- fg → text color
- font=("FontName", size) sets font style.

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10. Using Frames to Group Widgets

A **Frame** acts as a container to group multiple widgets.

```
import tkinter as tk

root = tk.Tk()
root.title("Frame Example")

frame = tk.Frame(root, bg="lightgray", bd=3)
frame.pack(padx=20, pady=20)

tk.Label(frame, text="This is inside the frame").pack()
tk.Button(frame, text="Click").pack()

root.mainloop()
```

Explanation:

- Frames help organize widgets neatly within sections.

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11. Displaying Images

Tkinter can display images using PhotoImage or the Pillow library.

Example with PNG image:

```
import tkinter as tk
```

```
root = tk.Tk()
root.title("Image Example")

photo = tk.PhotoImage(file="photo.png")
label = tk.Label(root, image=photo)
label.image = photo  # Prevent garbage collection
label.pack()

root.mainloop()
```

Example with JPG using Pillow:

```
from PIL import Image, ImageTk
import tkinter as tk

root = tk.Tk()
root.title("Image Example (JPG)")

img = Image.open("photo.jpg")
photo = ImageTk.PhotoImage(img)

label = tk.Label(root, image=photo)
label.image = photo
label.pack()

root.mainloop()
```

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12. Handling Mouse and Keyboard Events

You can make your app respond when the user types or clicks.

```
import tkinter as tk

def on_click(event):
    print("Mouse clicked at:", event.x, event.y)
```

```
root = tk.Tk()
root.title("Event Example")

root.bind("<Button-1>", on_click)  # Left mouse click
root.mainloop()
```

Explanation:

- .bind("<Button-1>", function) runs when the left mouse button is clicked.
- The event object gives info like x, y coordinates.

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13. Student Information Form Project

This small app takes user input and displays it back on screen.

```
import tkinter as tk

def show_info():
    name_val = name.get()
    age_val = age.get()
    class_val = cls.get()

    info = f"Name: {name_val}, Age: {age_val}, Class: {class_val}"
    result_label.config(text=info, fg="blue")

root = tk.Tk()
root.title("Student Information Form")
root.geometry("300x200")

tk.Label(root, text="Name:").grid(row=0, column=0, padx=5, pady=5)
name = tk.Entry(root)
name.grid(row=0, column=1, padx=5, pady=5)

tk.Label(root, text="Age:").grid(row=1, column=0, padx=5, pady=5)
```

```
age = tk.Entry(root)
age.grid(row=1, column=1, padx=5, pady=5)

tk.Label(root, text="Class:").grid(row=2, column=0, padx=5, pady=5)
cls = tk.Entry(root)
cls.grid(row=2, column=1, padx=5, pady=5)

tk.Button(root, text="Submit", command=show_info).grid(row=3, column=0, columnspan=2, pady=10)
result_label = tk.Label(root, text="")
result_label.grid(row=4, column=0, columnspan=2)

root.mainloop()
```

Explanation:

- Takes name, age, and class as input.
- Displays all entered details when “Submit” is clicked.

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14. Summary Table

Purpose	**Widget / Function**	**Example**
Text display	Label	Label(root, text="Hi")
Input box	Entry	Entry(root)
Button	Button	Button(root, text="Click")
Pop-up	messagebox	messagebox.showinfo()
Image	PhotoImage / ImageTk	Label(root, image=photo)
Layout	pack, grid, place	widget.grid()
Grouping	Frame	Frame(root)
Event	.bind()	root.bind("<Key>", func)
